Refraction for identifying matter (property of matter):

$$\begin{aligned} \eta &\overset{opt.}{=} \frac{sin(\alpha)}{\sin(\beta)} = \frac{\eta_\beta}{\eta_\alpha} \overset{prism}{=} \frac{\sin[\frac{1}{2}(\varpi + \varsigma_{min})]}{\sin[\frac{1}{2}\varpi]} \\ &\overset{phys.}{=} \sqrt{\epsilon_r \mu_r} = \frac{\lambda_\alpha}{\lambda_\beta} = \frac{v_{ph,\alpha}}{v_{ph,\beta}} = \frac{1}{2}\frac{\lambda}{d_l}\frac{n}{\cos(\beta)} = \sqrt{\left(\frac{n}{2}\frac{\lambda}{d_l}\right)^2 + \sin(\alpha)^2} \\ &\overset{dispersion}{=} \eta(\lambda) = \eta(\omega) \overset{anisotropy}{=} \eta(r) \end{aligned} \tag{1}$$

Vacuum $\iff \eta = 1$: [1]

$$f = FocalLength \qquad D = RefractivePower(dioptry) \tag{2}$$

$n \equiv$ order ("quantum"):

$$n = \frac{\nu}{c}\Delta s = \frac{\Delta\varphi}{2\pi} = 2\eta\frac{d_l}{\lambda}\sqrt{\eta^2 - \sin(\alpha)^2} = 2\eta\frac{d_l}{\lambda}\cos(\beta) = \frac{q}{e_0} \qquad \in \mathbb{N} \tag{3}$$

$$v^2 = \left(\frac{\nu\lambda}{\eta}\right)^2 = \left(\frac{\omega}{k\eta}\right)^2 \tag{4}$$

$$\left(\frac{\eta}{c}\right)^2 = \epsilon\mu \qquad \epsilon = \epsilon_0\epsilon_r \qquad \mu = \mu_0\mu_r \tag{5}$$

where:

$d_l :=$ distance length

$\alpha :=$ irradiation angle, incoming radiation direction, from source/sender (Einstrahlungswinkel zur Normalen auf Grenzfläche des Mediums)

$\beta :=$ refraction angle for bented transmitted radiation A_t orientation in matter (Durchstrahlungsswinkel zur Normalen aus Grenzfläche ins Medium)

[1]Brechkraft

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